Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Durrer Dairy

Physical address of dairy:

8861 Beckwith RD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 03/01/1980

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0012-0020-0011-0000

B. OPERATORS

Durrer, Chris

Operator name: Durrer, Chris

Telephone no.: (209) 545-1731 (Landline) (209) 595-5315 (Cellular)

8461 Beckwith RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

Durrer, Leo

Operator name: Durrer, Leo

Telephone no.: (209) 545-1731 (Landline)

8461 Beckwith RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

C. OWNERS

Durrer, Chris

Legal owner name:	Durrer, Chris	Telephone no.:	(209) 545-1731	(209) 595-5315
			Landline	Cellular
8461 Beckwith RD	Modesto		CA	95358
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

Durrer, Leo

Legal owner name:	Durrer, Leo	Telephone no.:	(209) 545-1731	
			Landline	Cellular
8461 Beckwith RD	Modesto		CA	95358
Mailing Address Number and Street	City		State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	75	280	240	100	100
Number under roof	495	0	0	0	0	0
Maximum number	495	75	280	240	100	100
Average number	495	75	280	240	100	100
Avg live weight (lbs)	1,400	1,450	1,100	900		

Predominant milk cow breed: Holstein

Average milk production: 86 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 20,870.95 *tons per reporting period*

Total nitrogen from manure: 251,421.57 *lbs per reporting period*

After ammonia losses (30% loss applied): 175,995.10 *lbs per reporting period*

Total phosphorus from manure: 41,220.27 *lbs per reporting period*

Total potassium from manure: 100,000.51 *lbs per reporting period*

Total salt from manure: 250,317.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 8,209,261 *gallons*

Total nitrogen generated: 24,767.75 *lbs*

Total phosphorus generated: 4,619.23 *lbs*

Total potassium generated: 35,250.95 *lbs*

Total salt generated: 300,350.71 *lbs*

	8,209,261	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	8,209,261	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal Sample	Surface water
Chris House	Ground water
Dad's House	Ground water
Milk Barn	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
East of Dairy & North of Dairy	90	90	2	process wastewater	0012-0020-0019-0000
Kellstrom	30	30	2	manure	0012-0020-0003-0000
Niemann	30	30	2	manure	0012-0020-0013-0000
Old Barn 1 & 2	75	75	2	manure	0012-0019-0008-0000
Totals for areas that were used for application	225	225	8		
Totals for areas that were not used for application					
Land application area totals	225	225	8		

B. CROPS AND HARVESTS

East of Dairy & North of Dairy

Field name: East of Dairy & North of Dairy

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 90 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	1,468.80 *ton*	Dry-weight		74.8	8,200.00	2,500.00	15,800.00		9.13

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	14.00
Total actual harvest content	16.32	67.45	20.56	129.96	750.97

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 90 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	2,232.90 *ton*	Dry-weight		58.4	14,200.00	3,100.00	14,700.00		5.37

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	28.00
Total actual harvest content	24.81	293.12	63.99	303.44	1,108.47

Kellstrom

Field name: Kellstrom

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	489.60 *ton*	Dry-weight		65.1	6,000.00	2,100.00	13,300.00		7.35

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	14.00
Total actual harvest content	16.32	68.35	23.92	151.51	837.26

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	744.30 *ton*	Dry-weight		65.6	10,600.00	2,300.00	9,200.00		4.14

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	28.00
Total actual harvest content	24.81	180.93	39.26	157.04	706.67

Niemann

Field name: Niemann

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	489.60 *ton*	Dry-weight		67.1	7,600.00	2,300.00	13,100.00		8.45

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	14.00
Total actual harvest content	16.32	81.61	24.70	140.68	907.41

Niemann

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	744.30 *ton*	Dry-weight		64.1	11,900.00	2,500.00	10,100.00		4.35

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	28.00
Total actual harvest content	24.81	211.98	44.53	179.92	774.89

Old Barn 1 & 2

Field name: Old Barn 1 & 2

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 75 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	1,224.00 *ton*	Dry-weight		56.4	13,700.00	3,400.00	29,100.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	14.00
Total actual harvest content	16.32	194.97	48.39	414.12	1,309.26

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	1,869.39 *ton*	Dry-weight		65.0	13,400.00	3,000.00	12,100.00		4.65

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	28.00
Total actual harvest content	24.93	233.80	52.34	211.12	811.32

NUTRIENT BUDGET

A. LAND APPLICATIONS

East of Dairy & North of Dairy - 10/28/2022: Oats, silage-soft dough

Field name: East of Dairy & North of Dairy

Crop: Oats, silage-soft dough — Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon application	Process wastewater	54.68	9.20	72.90	554.00	1,560,000.00 *gal*
Application event totals		54.68	9.20	72.90	554.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	27.51	4.83	39.66	325.12	720,000.00 *gal*
Application event totals		27.51	4.83	39.66	325.12	

East of Dairy & North of Dairy - 06/15/2023: Corn, silage

Field name: East of Dairy & North of Dairy

Crop: Corn, silage — Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	62.24	10.94	89.73	735.70	1,629,257.14 *gal*
Canal Sample	Surface water	0.10	0.00	0.00	21.62	11,105,016.68 *gal*
Application event totals		62.34	10.94	89.73	757.33	

East of Dairy & North of Dairy - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	47.17	9.51	68.31	621.29	1,551,052.80 *gal*
Canal Sample	Surface water	0.10	0.00	0.00	21.62	11,105,016.68 *gal*
Application event totals		47.27	9.51	68.31	642.91	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	45.98	9.27	66.59	605.63	1,511,950.63 *gal*
Canal Sample	Surface water	0.11	0.00	0.00	24.02	12,336,735.08 *gal*
Application event totals		46.10	9.27	66.59	629.65	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	14.90	3.00	21.58	196.27	490,000.00 *gal*
Canal Sample	Surface water	0.11	0.00	0.00	24.02	12,336,735.08 *gal*
Application event totals		15.02	3.00	21.58	220.30	

09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	9.91	2.00	14.36	130.58	326,000.00 *gal*
Canal Sample	Surface water	0.09	0.00	0.00	19.45	9,986,694.58 *gal*
Application event totals		10.01	2.00	14.36	150.03	

09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	12.80	2.58	18.54	168.64	421,000.00 *gal*
Canal Sample	Surface water	0.09	0.00	0.00	19.45	9,986,694.58 *gal*
Application event totals		12.90	2.58	18.54	188.08	

Kellstrom - 10/28/2022: Oats, silage-soft dough

Kellstrom - 10/28/2022: Oats, silage-soft dough

Field name: Kellstrom

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Spread Solid Manure	Separator solids	72.77	18.56	15.14	0.00	111.00 *ton*
Application event totals		72.77	18.56	15.14	0.00	

Kellstrom - 06/15/2023: Corn, silage

Field name: Kellstrom

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.10	0.00	0.00	21.62	3,701,672.23 *gal*
Application event totals		0.10	0.00	0.00	21.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Spread Solid Manure	Separator solids	182.19	46.46	37.91	0.00	277.90 *ton*
Application event totals		182.19	46.46	37.91	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.10	0.00	0.00	21.62	3,701,672.23 *gal*
Application event totals		0.10	0.00	0.00	21.62	

Kellstrom - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.10	0.00	0.00	21.62	3,701,672.23 *gal*
Application event totals		0.10	0.00	0.00	21.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.10	0.00	0.00	21.62	3,701,672.23 *gal*
Application event totals		0.10	0.00	0.00	21.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.09	0.00	0.00	19.45	3,328,898.19 *gal*
Application event totals		0.09	0.00	0.00	19.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.09	0.00	0.00	19.45	3,328,898.19 *gal*
Application event totals		0.09	0.00	0.00	19.45	

Niemann - 10/28/2022: Oats, silage-soft dough

Field name: Niemann

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Spread Solid Manure	Separator solids	72.77	18.56	15.14	0.00	111.00 *ton*
Application event totals		72.77	18.56	15.14	0.00	

Niemann - 06/15/2023: Corn, silage

Field name: Niemann

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.10	0.00	0.00	21.62	3,701,672.23 *gal*
Application event totals		0.10	0.00	0.00	21.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Spread Solid Manure	Separator solids	247.75	63.18	51.55	0.00	377.90 *ton*
Application event totals		247.75	63.18	51.55	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.10	0.00	0.00	21.62	3,701,672.23 *gal*
Application event totals		0.10	0.00	0.00	21.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.10	0.00	0.00	21.62	3,701,672.23 *gal*
Application event totals		0.10	0.00	0.00	21.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.10	0.00	0.00	21.62	3,701,672.23 *gal*
Application event totals		0.10	0.00	0.00	21.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.09	0.00	0.00	19.45	3,328,898.19 *gal*
Application event totals		0.09	0.00	0.00	19.45	

Niemann - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.09	0.00	0.00	19.45	3,328,898.19 *gal*
Application event totals		0.09	0.00	0.00	19.45	

Old Barn 1 & 2 - 10/28/2022: Oats, silage-soft dough

Field name: Old Barn 1 & 2

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Spread Solid Manure	Separator solids	72.77	18.56	15.14	0.00	277.50 *ton*
Application event totals		72.77	18.56	15.14	0.00	

Old Barn 1 & 2 - 06/15/2023: Corn, silage

Field name: Old Barn 1 & 2

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.10	0.00	0.00	21.62	9,254,180.57 *gal*
Application event totals		0.10	0.00	0.00	21.62	

Old Barn 1 & 2 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Spread Solid Manure	Separator solids	302.82	77.23	63.00	0.00	1,154.75 *ton*
Application event totals		302.82	77.23	63.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.10	0.00	0.00	21.62	9,254,180.57 *gal*
Application event totals		0.10	0.00	0.00	21.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.10	0.00	0.00	21.62	9,254,180.57 *gal*
Application event totals		0.10	0.00	0.00	21.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.10	0.00	0.00	21.62	9,254,180.57 *gal*
Application event totals		0.10	0.00	0.00	21.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.09	0.00	0.00	19.45	8,322,245.48 *gal*
Application event totals		0.09	0.00	0.00	19.45	

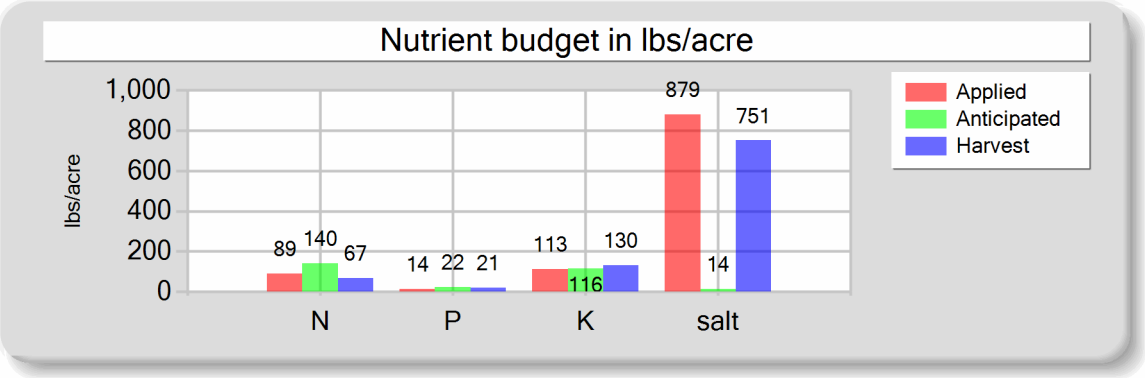
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Sample	Surface water	0.09	0.00	0.00	19.45	8,322,245.48 *gal*
Application event totals		0.09	0.00	0.00	19.45	

B. NUTRIENT BUDGET

East of Dairy & North of Dairy - 10/28/2022: Oats, silage-soft dough

Field name: East of Dairy & North of Dairy Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	82.18	14.03	112.56	879.12
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	89.18	14.03	112.56	879.12
Anticipated crop nutrient removal	140.00	22.40	116.20	14.00
Actual crop nutrient removal	67.45	20.56	129.96	750.97
Nutrient balance	21.73	-6.53	-17.40	128.15
Applied to removed ratio	1.32	0.68	0.87	1.17

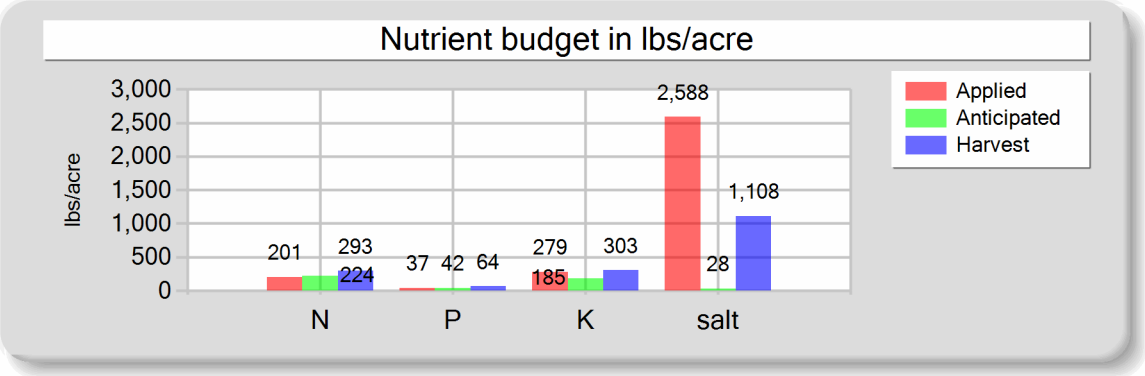
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
2,280,000.00 *gallons*
83.96 *acre-inches*
0.93 *inches/acre*

Total harvests for the crop
1 *harvests*

East of Dairy & North of Dairy - 06/15/2023: Corn, silage

Field name: East of Dairy & North of Dairy Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	193.02	37.29	279.12	2,458.11
Fresh water	0.62	0.00	0.00	130.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	200.64	37.29	279.12	2,588.29
Anticipated crop nutrient removal	224.00	42.00	184.80	28.00
Actual crop nutrient removal	293.12	63.99	303.44	1,108.47
Nutrient balance	-92.48	-26.70	-24.32	1,479.82
Applied to removed ratio	0.68	0.58	0.92	2.34

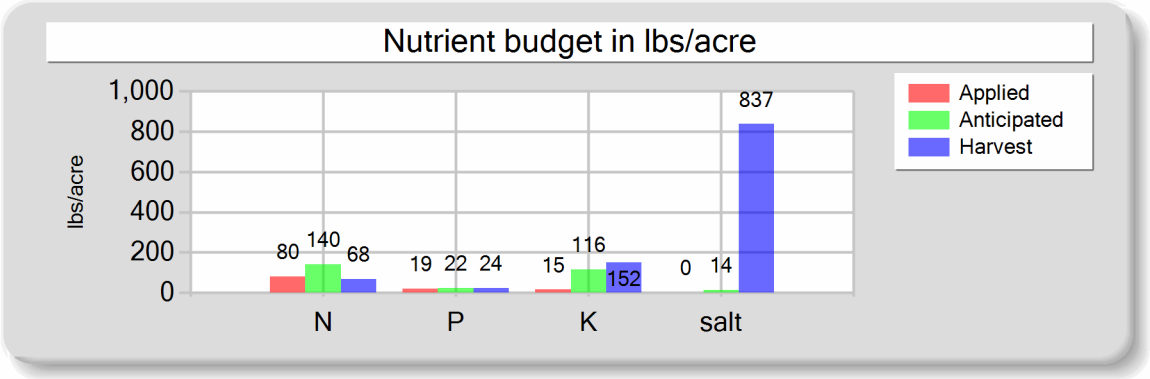
Fresh water applied	
66,856,892.68	*gallons*
2,462.11	*acre-inches*
27.36	*inches/acre*

Process wastewater applied	
5,929,260.57	*gallons*
218.35	*acre-inches*
2.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

Kellstrom - 10/28/2022: Oats, silage-soft dough

Field name: Kellstrom Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	72.77	18.56	15.14	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	79.77	18.56	15.14	0.00
Anticipated crop nutrient removal	140.00	22.40	116.20	14.00
Actual crop nutrient removal	68.35	23.92	151.51	837.26
Nutrient balance	11.42	-5.36	-136.36	-837.26
Applied to removed ratio	1.17	0.78	0.10	0.00

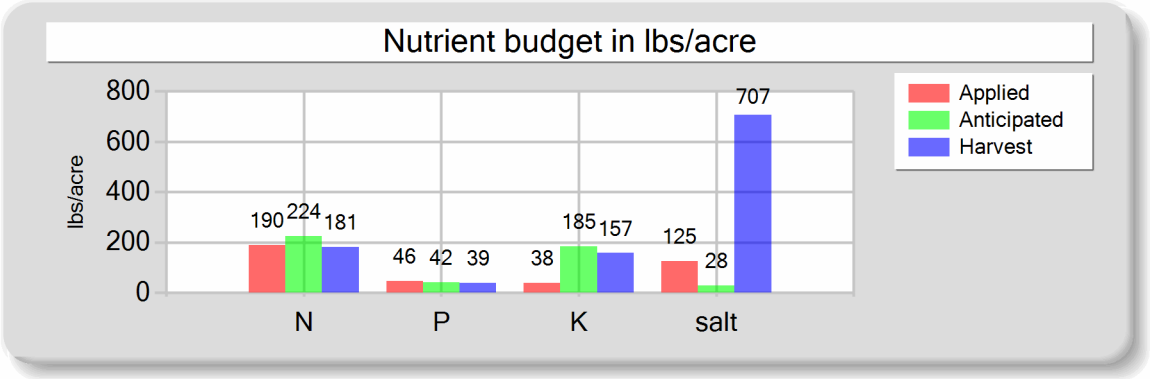
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Kellstrom - 06/15/2023: Corn, silage

Field name: Kellstrom Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	182.19	46.46	37.91	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.60	0.00	0.00	125.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	189.79	46.46	37.91	125.38
Anticipated crop nutrient removal	224.00	42.00	184.80	28.00
Actual crop nutrient removal	180.93	39.26	157.04	706.67
Nutrient balance	8.85	7.21	-119.13	-581.28
Applied to removed ratio	1.05	1.18	0.24	0.18

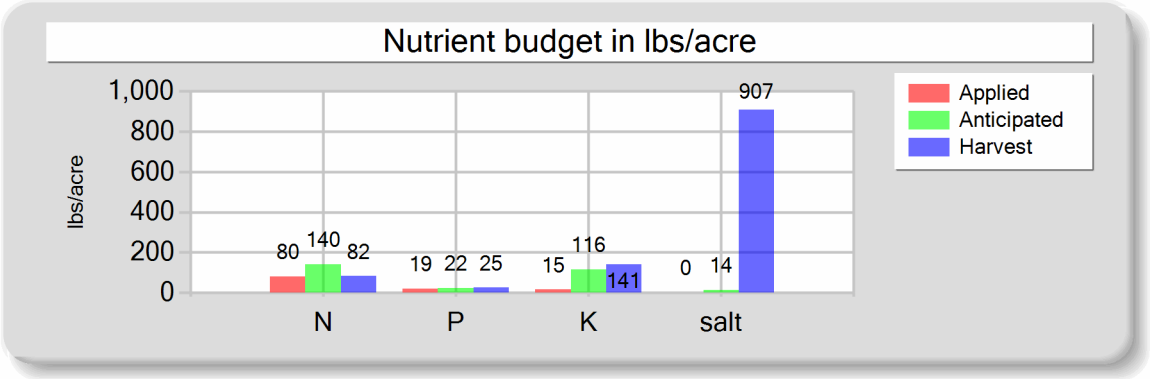
Fresh water applied	
21,464,485.30	*gallons*
790.46	*acre-inches*
26.35	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Niemann - 10/28/2022: Oats, silage-soft dough

Field name: Niemann | Crop: Oats, silage-soft dough | Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	72.77	18.56	15.14	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	79.77	18.56	15.14	0.00
Anticipated crop nutrient removal	140.00	22.40	116.20	14.00
Actual crop nutrient removal	81.61	24.70	140.68	907.41
Nutrient balance	-1.84	-6.14	-125.53	-907.41
Applied to removed ratio	0.98	0.75	0.11	0.00

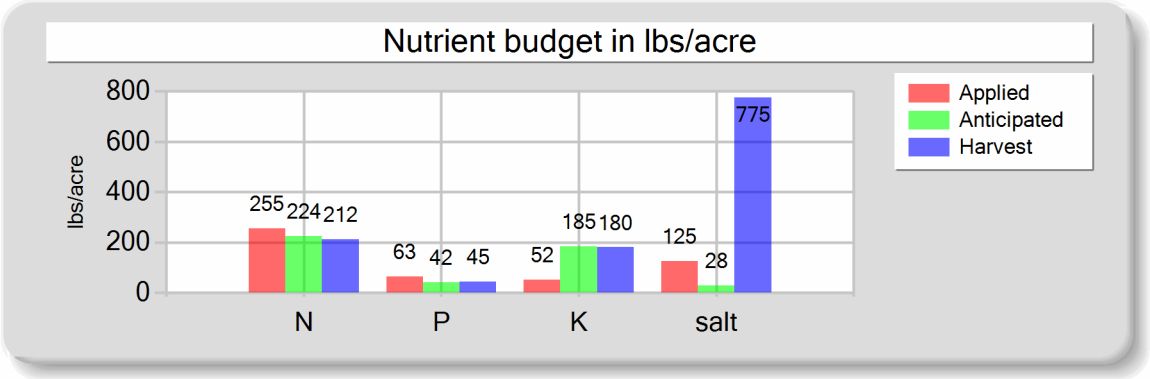
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Niemann - 06/15/2023: Corn, silage

Field name: Niemann Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	247.75	63.18	51.55	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.60	0.00	0.00	125.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.35	63.18	51.55	125.38
Anticipated crop nutrient removal	224.00	42.00	184.80	28.00
Actual crop nutrient removal	211.98	44.53	179.92	774.89
Nutrient balance	43.37	18.65	-128.37	-649.51
Applied to removed ratio	1.20	1.42	0.29	0.16

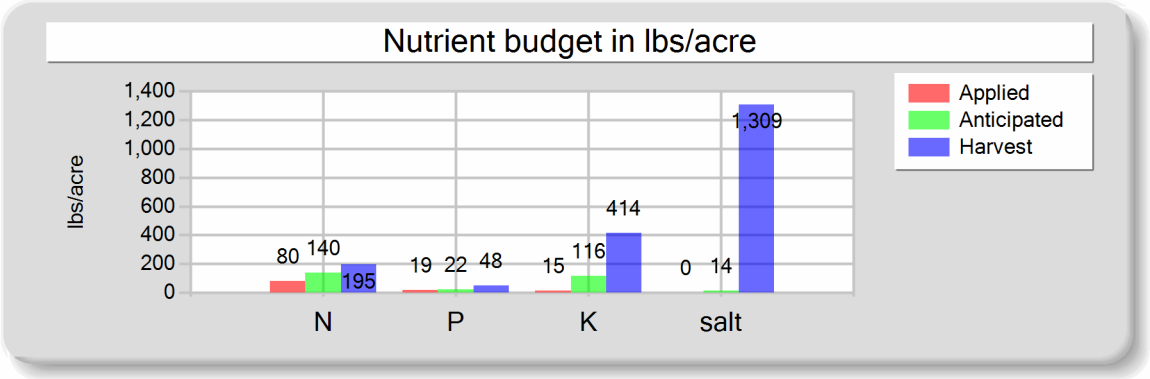
Fresh water applied	
21,464,485.30	*gallons*
790.46	*acre-inches*
26.35	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Old Barn 1 & 2 - 10/28/2022: Oats, silage-soft dough

Field name: Old Barn 1 & 2 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	72.77	18.56	15.14	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	79.77	18.56	15.14	0.00
Anticipated crop nutrient removal	140.00	22.40	116.20	14.00
Actual crop nutrient removal	194.97	48.39	414.12	1,309.26
Nutrient balance	-115.19	-29.83	-398.98	-1,309.26
Applied to removed ratio	0.41	0.38	0.04	0.00

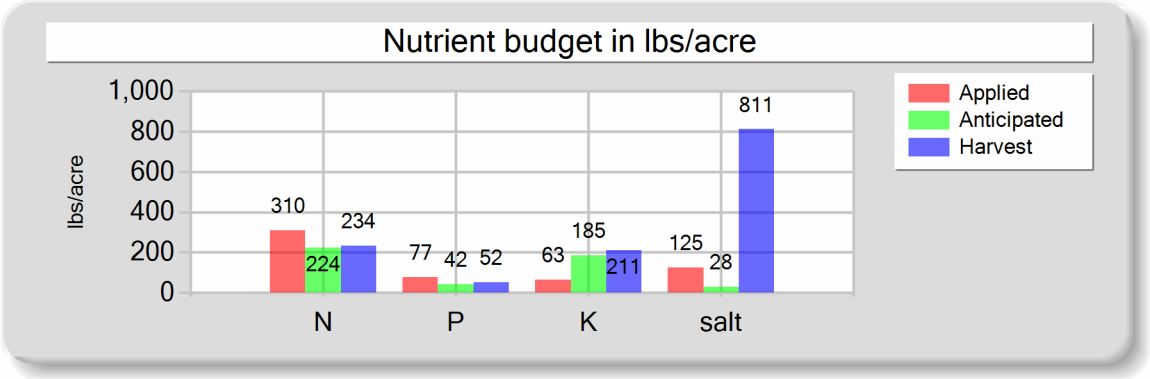
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Old Barn 1 & 2 - 06/15/2023: Corn, silage

Field name: Old Barn 1 & 2 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	302.82	77.23	63.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.60	0.00	0.00	125.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	310.42	77.23	63.00	125.38
Anticipated crop nutrient removal	224.00	42.00	184.80	28.00
Actual crop nutrient removal	233.80	52.34	211.12	811.32
Nutrient balance	76.62	24.89	-148.11	-685.93
Applied to removed ratio	1.33	1.48	0.30	0.15

Fresh water applied
53,661,213.24 *gallons*
1,976.16 *acre-inches*
26.35 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Spring Corral Manure

Sample and source description: Spring Corral Manure

Sample date: 06/23/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 94.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,200.00	8,900.00	16,900.00							
DL	0.01	0.01	0.01							

Spring Separator Manure

Sample and source description: Spring Separator Manure

Sample date: 06/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 34.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	3,800.00	3,100.00							
DL	0.01	0.01	0.01							

B. PROCESS WASTEWATER ANALYSES

1st Quarter Lagoon

Sample and source description: 1st Quarter Lagoon

Sample date: 04/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	412.00	321.00	0.00	0.00	72.40	594.00								7,330.00	4,870
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

2nd Quarter Lagoon

Sample and source description: 2nd Quarter Lagoon

Sample date: 06/23/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	328.00	283.00	0.00	0.00	66.10	475.00								6,510.00	4,320
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

3rd Quarter Lagoon

Sample and source description: 3rd Quarter Lagoon

Sample date: 08/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	430.00	283.00	0.00	0.00	58.00	620.00								7,120.00	6,540
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

4th Quarter Lagoon

Sample and source description: 4th Quarter Lagoon

Sample date: 10/17/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	378.00	268.00	0.00	0.00	63.60	504.00								5,770.00	3,830
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

C. FRESH WATER ANALYSES

Canal Sample

Canal Sample

Canal Sample

Sample description: Canal Sample

Sample date: 09/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.10		0.10								30.00	21
DL	0.01		0.01								0.01	1

Chris House

Domestic Well

Sample description: Domestic Well

Sample date: 06/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	20.20		20.20								1,220.00	778
DL	0.01		0.01								0.01	1

Dad's House

Domestic Well

Sample description: Domestic Well

Sample date: 06/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	20.30		20.30								448.00	287
DL	0.01		0.01								0.01	1

Milk Barn

Milk Barn

Domestic Well

Sample description: Domestic Well

Sample date: 06/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	81.00		81.00								1,890.00	1,210
DL	0.01		0.01								0.01	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

East of Dairy & North of Dairy - 10/28/2022: Oats, silage-soft dough

Winrter Forage

Sample and source description: Winrter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,200.00	2,500.00	15,800.00		9.13
DL	0.01	0.01	0.01		1.00

East of Dairy & North of Dairy - 06/15/2023: Corn, silage

East of Dairy & North of Dairy - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 58.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	3,100.00	14,700.00		5.37
DL	0.01	0.01	0.01		1.00

Corn Silage

Sample and source description: Corn Silage
Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,300.00	6,400.00		4.49
DL	0.01	0.01	0.01		1.00

Kellstrom - 10/28/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,000.00	2,100.00	13,300.00		7.35
DL	0.01	0.01	0.01		1.00

Kellstrom - 06/15/2023: Corn, silage

Kellstrom - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	2,300.00	9,200.00		4.14
DL	0.01	0.01	0.01		1.00

Niemann - 10/28/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,600.00	2,300.00	13,100.00		8.45
DL	0.01	0.01	0.01		1.00

Niemann - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,900.00	2,500.00	10,100.00		4.35
DL	0.01	0.01	0.01		1.00

Old Barn 1 & 2 - 10/28/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	3,400.00	29,100.00		9.20
DL	0.01	0.01	0.01		1.00

Old Barn 1 & 2 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	3,000.00	12,100.00		4.65
DL	0.01	0.01	0.01		1.00

Corn Silage

Sample and source description: Corn Silage

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	2,900.00	9,400.00		5.44
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

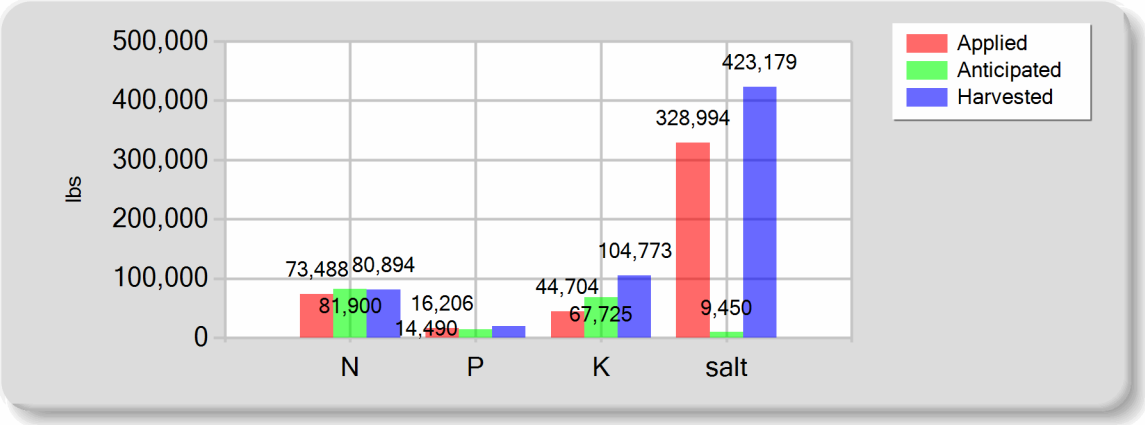
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

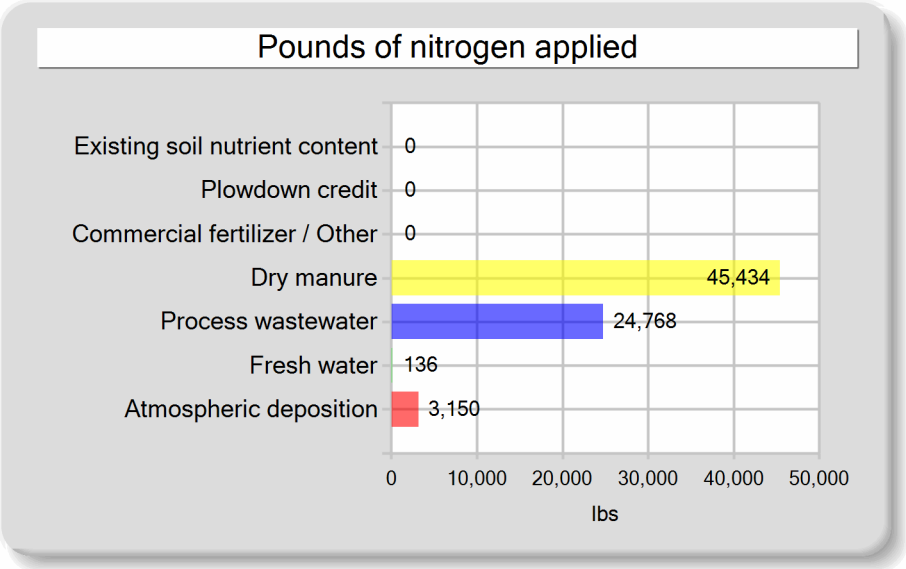
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

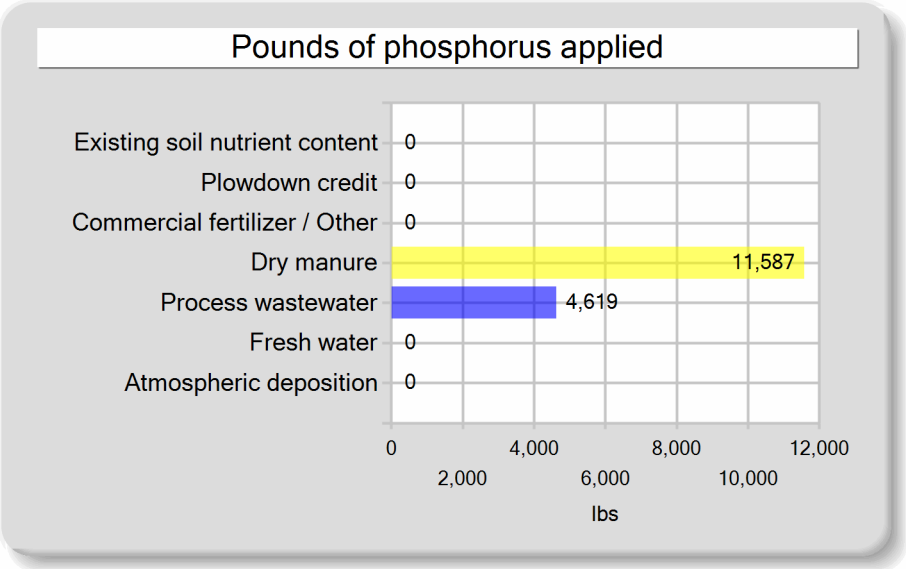
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	45,434.06	11,587.21	9,452.72	0.00
Process wastewater	24,767.75	4,619.23	35,250.95	300,350.71
Fresh water	136.40	0.00	0.00	28,643.28
Atmospheric deposition	3,150.00	0.00	0.00	0.00
Total nutrients applied	73,488.21	16,206.44	44,703.68	328,994.00
Anticipated crop nutrient removal	81,900.00	14,490.00	67,725.00	9,450.00
Actual crop nutrient removal	80,894.22	19,136.83	104,772.63	423,179.31
Nutrient balance	-7,406.01	-2,930.39	-60,068.95	-94,185.32
Applied to removed ratio	0.91	0.85	0.43	0.78

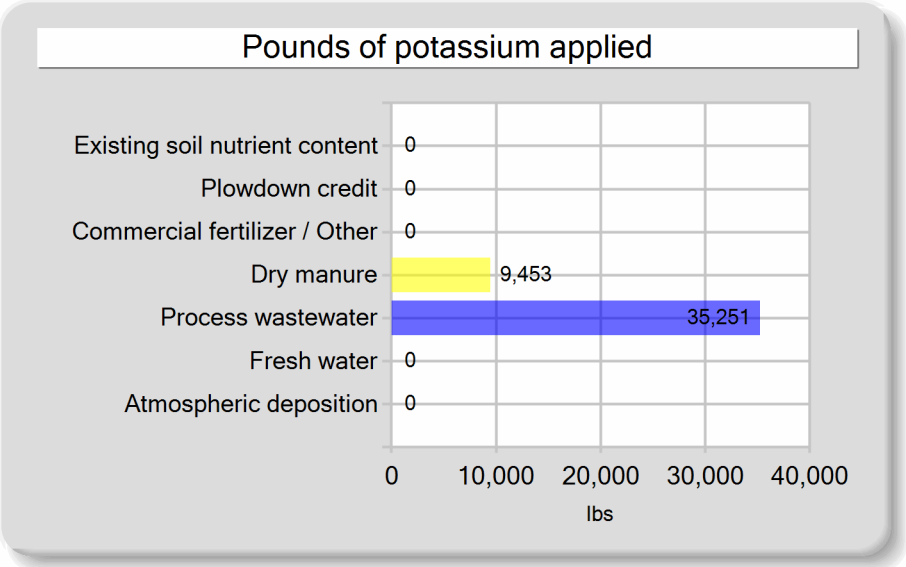
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

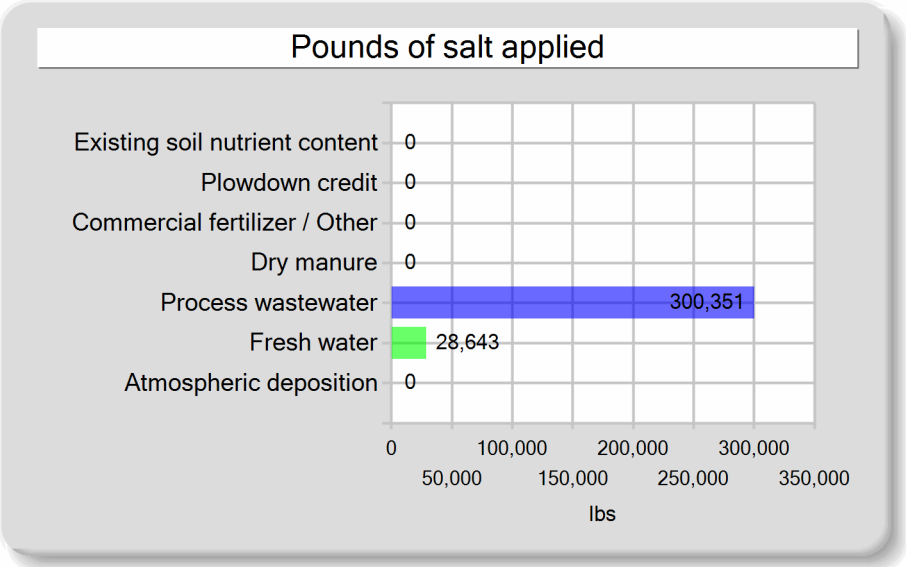


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Chris Durrer	Chris Durrer
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Chris Durrer	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-27-24	6-27-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
DURRER DAIRY LP
8461 Beckwith Rd
Modesto, CA 95358

Grower

Consultant:

PURCHASE ORDER:
RECEIVED DATE: 6/28/23 10:15 AM
SUBMITTED BY: Jenny
ANALYZED DATE: 6/28/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Chris House	T3179007A	Electrical Conductivity (EC)	EPA 120.1 and	1.22	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	20.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	778	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	102	94.6 (T3178005A)	93.9 (T3178005A)	0.699
Electrical Conductivity (EC)	99.5			0.578
Nitrate Nitrogen (NO3-N)	102	106 (T3179012A)	106 (T3179012A)	0.453

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
DURRER DAIRY LP
8461 Beckwith Rd
Modesto, CA 95358

Grower

Consultant:

PURCHASE ORDER:
RECEIVED DATE: 6/28/23 10:15 AM
SUBMITTED BY: Jenny
ANALYZED DATE: 6/28/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dad's House	T3179007C	Electrical Conductivity (EC)	EPA 120.1 and	0.448	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	20.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	287	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	102	94.6 (T3178005A)	93.9 (T3178005A)	0.699
Electrical Conductivity (EC)	99.5			0.578
Nitrate Nitrogen (NO3-N)	102	106 (T3179012A)	106 (T3179012A)	0.453

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Purrer Dairy
Mailing address:
(not facility address)
Phone#:

Author: Central Valley Ag Consulting
Jenny Mesenhimer-Martin
centralvalleyag@yahoo.com
(209) 480-1757
Sampl
Projec

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	6-28-23		Chris House		21.0°		X				
2											
3			Milk Barn		21.0°		X				
4											
5			Dad's House		21.0°		X				
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st	[signature]	CVAC	10:15 6/28/23			
2nd	[signature]	DENELE ANALYTICAL, INC.	6-28-23 10:15			T 3179007

(LABORATORY USE ONLY)

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ANALYZED DATE: 6/28/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Milk Barn	T3179007B	Electrical Conductivity (EC)	EPA 120.1 and	1.89	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	81.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1210	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	102	94.6 (T3178005A)	93.9 (T3178005A)	0.699
Electrical Conductivity (EC)	99.5			0.578
Nitrate Nitrogen (NO3-N)	102	106 (T3179012A)	106 (T3179012A)	0.453

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